

Agostinho A. Soto-wang

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Summary

An aspiring translational immunological researcher, fascinated by driving action through multimodal data. Passionate about implementing machine learning/data algorithms to explain biological phenomena.

Education

UC San Diego

B.S, Bioengineering: Bioinformatics

- 3.90 GPA

Expected June 2029

Experience

Ay Lab – La Jolla Institute of Immunology, La Jolla, CA

08/2026-Current

Undergraduate Research Assistant

- Analyzing scRNA-seq data for differential transposable element expression in patients

MIND Lab – UC San Diego, La Jolla, CA

01/2026-08/2026

Volunteer Lab Research Assistant

- Assessed spatial-temporal feature importance for tracked immune cells through LSTM-VAE structure
 - Automated microscopy LIF file processing, reducing time demand by 30%
 - Trained open-source image segmentation model for identifying spheroids in brightfield microscopy
 - Applied dimensionality reduction (UMAP, PCA) to csv data to identify and represent cell subpopulations
 - Utilized Rheometer for viscoelastic characterization of alginate-collagen hydrogels
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Projects/Research

Kaggle RSNA Knee Abnormality Detection Competition

08/2026 - Present

- Employing HuggingFace MedicalBERT text transformer to generate embeddings from medical reports
- Utilizing Vit3D transformer to analyze 3D DICOM MRI scans
- Working towards creating shared embedding space for model interpretability across data modalities

Kaggle Biohub Cell Tracking Competition

07/2026 - Present

- Deployed 3D UNET model for identifying zebrafish embryos from timelapse microscopy
- Employed dual-decoder structure to return embryo and division probability mesh grid
- Filtered image through watershed segmentation and Otsu thresholding

Immune Cell Infiltration Prediction

01/2026 – 08/2026

- Developed VAE-LSTM structure for exploring latent representation of single cell morphological and migratory features
 - Employed free-bits and cyclical beta-annealing to improve decoder reconstructions 10%
 - Established Logistic Regression model as baseline, achieving ROC-AUC of 0.80
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SKILLS & ACTIVITIES

- Libraries: Pandas, Numpy, SQL, Matplotlib, PyTorch, Tensorflow, Scikit-learn, OpenCV, Scanpy, AnnData
- Systems: Python, Linux, Git, AWS, HuggingFace, Nextflow, Docker
- Biomedical Engineering Society: Lab Expo Chair and Outreach High School Liaison